KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

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DEPARTMENT OF MATHEMATICS

ELLIPTIC CURVE CRYPTOGRAPHY

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A THESIS SUBMITTED TO THE DEPARTMENT OF MATHEMATICS, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BSc MATHEMATICS

Declaration

We hereby conscientiously declare that this dissertation is our own work under the able supervision of Dr. Richard K. Boadi, towards the award of the Bachelor of Science(BSc.) in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which had been accepted for the award of any other degree of the university, except where due acknowledgement had been made in the text.

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Dedication

We humbly dedicate this work to the Almighty God for His grace, mercy and favour throughout our four-year academic endurance. We are not self made, our reaching this point in our academic life has been by the collective effort of some family and friends and so we would like to dedicate this work to every single individual who has contributed invaluably in one way or the other to our reaching this point in life.

Acknowledgement

We are most grateful to the Almighty God for His mercies and guidance. For His Grace by which we can still hope.We are profoundly grateful to Dr.Richard K. Boadi (our project supervisor).We would like to show great appreciation for his devotion, guidance, support and encouragement.

We are truly indebted to Wofa Kwabena Adu-Gyamfi for the enormous amount of time he spent to provide guidance, assistance and helpful constructive critique to our work.And to all the lecturers in the Department of Mathematics, KNUST, we say God bless you.

Abstract

Koblitz and Miller introduced the use of elliptic curves in encryption in 1985. The function of elliptic curves in cryptography was investigated.In this thesis we explored the basic implementation and security issues of several elliptic curve cryptosystems.Some of the aspect that will be addressed includes;Using a single solution to obtain many elliptic curve solutions,finding points algebraically,Application of rational numbers on elliptic curves in cryptography and how well programming codes are used to find private key cryptography.

Contents

1 INTRODUCTION **1**

1.1 Background of the study 1

1.2 Problem Statement 2

1.3 Objectives of study 3

1.4 Significance of the study 3

1.5 Scope of the study 3

1.6 Organization of Study 4

2 LITERATURE REVIEW **5**

3 Methodology **9**

3.1 Group Law 9

3.1.1 Weierstrass Equation 9

3.1.2 Group Properties 10

3.2 Using one solution to find various solution to an elliptic curve 11

3.3 Finding point algebraically 13

3.3.1 Identity 15

3.3.2 Inverse 15

3.3.3 Associative 16

3.3.4 Point Doubling 18

3.4 cyclic group 20

3.4.1 The Key Pair Generation of ECC 20

3.5 Diffie - Hellman Key Exchange on elliptic curve cryptography 20

3.6 ElGamal Public Key Encryption 21

3.7 ElGamal Digital Signatures 22

4 . . . . . **24**

4.1 Application of rational numbers on elliptic curves in cryptography. 24

4.1.1 Fermat curves 24

4.1.2 The Mordell Weil group 25

4.2 The Discrete Logarithm Problems 25
4.3 Attacks on Discrete logs 26
4.3.1 Attacks Aiming to Recover x 26
4.3.2 Attacks for Forging Signatures 27
4.3.3 Baby Step, Giant Step 28
4.3.4 Pollard's ρ Method 29
4.4 Elliptic Curve Domain Parameters 32
4.4.1 Elliptic Curve Domain Parameters over Fp 32
4.4.2 Recommended Parameters secp192k1 32

5 Conclusion and Recommendation 36
5.1 Conclusion 36
5.2 Recommendation 36

References 38

Chapter 1

INTRODUCTION

Elliptic curve cryptography is a type of public-key encryption used in institutions and government computer system. The Diffie-Hellman key exchange enables the transfer of secure information on the ElGamal public key exchange. ElGamal digital signature ensures the authentication of information among individuals. To high-light the aforementioned, it is essential to understand attacks on discrete logs, this will help to maximize existing approaches to maintaining security. Using a recommended domain parameters of secp192k1, we generate a private key.

1.1 Background of the study

Manoj (2010) defined cryptography as the art and science of hiding messages in order to add secrecy to information security. The term "Cryptography" was coined by combining two Greek words; "Krypto," which means "hidden," and "Graphene," which means "writing," which is where cryptography is thought to have begun. As civilizations advanced, tribes, clans, and kingdoms arose. This resulted in the emergence of power, disputes, supremacy, and politics. These ideas encouraged people's natural need to engage privately with a limited group of people, guaranteeing that cryptography progressed.

The beginning of cryptography can be traced back to the Roman and Egyptian civilizations. Hieroglyph is the oldest cryptographic technique. The first documented hint of cryptography is the use of the term 'Hieroglyph'. Around 4000 years ago, the Egyptians used hieroglyphic messages to communicate. This code was only known by the scribes who used to deliver communications on behalf of the rulers.

Years later, in 1976, Diffie and Hellman invented public key cryptography, which not only revitalized the field of cryptography, but also influenced the direction of computational number theory research. For the first time, the relative difficulty of various number theorem tasks became a reasonable concern.

The rivest-shamir-adleman (RSA) cryptosystem was the first feasible public key system, having been introduced in 1978. It spurred a flurry of research into factoring and primality testing, two related issues. A type of public key cryptography based on the discrete counterpart of the

logarithm function spawned a second line of research in computational number theory.
discrete log cryptosystems (DLC) have not been around as long as RSA, but they have caught academics interest. The structure has prompted a lot of research due to the practical challenges that have arisen in discrete log cryptography.

In Afreen and Mehrotra (2011) Neal Koblitz and Victor Miller independently proposed the usefulness of elliptic curves in encryption in 1985. Since then, elliptic curve cryptography (ECC) has grown into a substantial branch of Public Key Encryption.

Public key cryptosystem (PKC) use public keys to encrypt data. Data is encrypted and decoded using different keys in a PKC system. Because one of the keys in a PKC system is made public and the key size determines the system's security.

Prior to PKC, mathematical problems were solved using prime factorization and discrete logarithm.

Elliptic curve cryptography has demonstrated that smaller key sizes can achieve the same level of security. For the most part, elliptic curve cryptography is concerned with the security of application systems, storage, computing performance, as well as CPU and GPU architecture.

Cryptography has been useful from the past and it is still now. Cryptography has undergone series of evolution that has led to cloud computing, data security issues, challenges in the cloud, DNA computing and DNA cryptography.
Cloud computing provide pervasive, easy, on-demand network connectivity, to a configurable pool of computational resources; of which include network infrastructures, servers, on & off storage sites, applications and services. All these can be accessed and released quickly and with low effort and interaction with service providers or management effort.

1.2 Problem Statement

Elliptic curve cryptography has many benefits that have helped in the field of science and technology in terms of information secrecy and keeping of confidential data for governments and institutions. Nevertheless, it does have its negative aspects. Channel attacks and twist-security attacks are examples of such attacks. Both types try to compromise the ECC's private key security. This project seeks to address and highlight some possible solutions to these problems.

1.3 Objectives of study

This project's major goal is to better understand elliptic curves and how they relate to cryptography. Abstract algebra concepts such as groups and fields are examined first as a prerequisite for the thesis.

Other Objectives are;

- To examine cryptographic applications of elliptic curves.
- To understand how private key is generated and the attacks on security.

1.4 Significance of the study

Elliptic curves can be found in a wide range of mathematical fields, from number theory to complex analysis. Elliptic curves can be used for encryption, digital signatures, pseudo-random generators, and a variety of other applications. They are also employed in a number of integer factorization methods with cryptographic applications, such as the Lenstra elliptic-curve factorization, illustrating the importance of elliptic curves in human lives in the 1980s and years after. This thesis would illustrate how to employ elliptic curves in various applications, particularly when they are defined over finite fields. It also examines how and why such curves are useful in the field of cryptography.

1.5 Scope of the study

This thesis is not intended to address a specific topic or to produce a novel finding; rather, as previously said, it intends to provide a grasp of elliptic curves, how they came to be, and how they are applied in elliptic curve cryptography. The subject of elliptic curves is a strong blend of abstract algebra, in particular group law and fields, as we will see in this paper.

At the moment, elliptic curves can be used for encryption, digital signatures, pseudo-random generators, and a variety of other applications. They are also employed in a number of integer factorization methods with cryptographic applications. Elliptic curves are employed in cryptographic protocols like Bitcoin and Austrian smart e-ID, to name a few examples. We will go

over each of these applications. Elliptic curves are a common theme in algebraic geometry and cryptography, among other branches of mathematics. We can find number theory problems in the history of elliptic curves and complex theoretical functions.

1.6 Organization of Study

This thesis is divided into five chapters, the first of which provides a broad summary of the entire thesis as well as the circumstances of the research. This chapter begins with an introduction, the study's history, a problem statement, and the study's goal. We outline the work connected to elliptic curve cryptography in the project's second chapter. The third chapter contains thorough information on elliptic curve cryptograph. Detailed results and analysis about the generation of private key cryptography are presented in chapter four. The conclusions and recommendations from the results and discussions have been outlined in chapter five.

Chapter 2

LITERATURE REVIEW

In this chapter, we summarize the works which are related to the case study of the elliptic curve cryptography.

A lot of research has been conducted in this field and we intern to review and analyze the information or data gathered so far.

Narayan (2018a) presented the order of the related algebraic group that was significantly easier to determine than in the general case for a family of elliptic curves for cryptographic purposes. This made it easier to meet the cryptographic requirement of a big prime factor for this arrangement. Another merit of this family was that group operations are slightly simplified.

Narayan (2018b) presented a detailed overview of cryptography method that was based on complex mathematical problems such as prime number factorization and the discrete logarithm problem for elliptic curves. The method of continuously adding a point to an elliptic curve was known as elliptic curve point multiplication. It is used to create a trapdoor function in elliptic curve cryptography (ECC).

Koblitz et al. (2000) looked over the growth of elliptic curve cryptosystems. In 1976 Diffie and Hellman introduced the public-key cryptography. It was found out that the discrete logarithm problem had the potential to be used in public-key cryptosystems. Diffie and Hellman described the discrete logarithm problem as the challenge of finding logarithms with regard to a generator in the multiplicative group of the integers modulo a prime, the discrete logarithm problem had since been redefined. This idea could be widened to arbitrary groups and, in particular, to elliptic curve groups. The resulting public-key systems had a small block size, are fast, and secure.

Koblitz (1987) discussed public key cryptosystems that used the multiplicative group of a finite field had elliptic curves over finite fields. This is because the elliptic curve equivalent of the discrete logarithm problem was anticipated to be harder than the conventional discrete logarithm problem, especially over $GF(2^n)$, these elliptic curve cryptosystems may be more safe on the nonsmoothness of the order of the cyclic subgroup created by a global point.

Smart (1999) proposed a basic technique that addressed the discrete logarithm issue on trace one elliptic curves, which led to a linear algorithm. In practice, this meant that when elliptic curves were selected for cryptography, all curves with group orders equal to the finite field's order must be eliminated.

Barbulescu et al. (2015) investigated a field $GF(p_n)$, where n was a significant integer bigger that 1, the hardness of the discrete logarithm issue was examined. The difficulty of this challenge was at the center of security evaluations for torus-based and pairing-based encryption, while being less studied than the tiny characteristic case or the prime field situation. The number field sieve(NFS) was the most well-known approach for tackling this problem. The ability to identify appropriate polynomials that specified the extension fields used in NFS was a significant component of this technique. Two new techniques for this problem was provided, each of which modifies the asymptotic complexity and allows for record-breaking calculations. It demonstrated these findings by computing discrete logarithms over a field $GF(p_2)$ with cardinality of $180digits(595bits)$. Chen (2008) observed how the Legendre symbol behaved when applied to rational points on an elliptic curve, and came up with a family of binary sequences that had string pseudorandom features. That was, both the well-distribution measure and the correlation measure of order k of the generated binary sequences are "small" as measured by estimate of particular character sums along elliptic curves. It also offered a lower constraint on the linear complexity profile of these sequences. The findings suggested that such sequences behave similarly to Legendre sequences.

Koblitz et al. (2000) reviewed the growth on elliptic curve cryptosystems. Elliptic curve cryptography has seen increasing success in security applications in recent years due to features such as a small key size with a high level of protection. Their popularity has resulted in a wide range of implementations in terms of algorithms, curves, coordinate systems, platforms, and so forth.

Vidhya et al. (2019) highlighted works that overcame the drawback of the GCD attack in RSA. One of the most important varieties of Asymmetric cryptography was Rivest, Shamir, and Adleman (RSA). When compared to other cryptography techniques, the ky size in RSA is larger. Any two prime numbers were used to create the keys, with the Greatest Common Divisor (GCD) attacks, an unauthorized user might simply obtain the keys . One of RSA cryptography's major flaws was with a small key size. Elliptic curve cryptography (ECC) is

also asymmetric cryptography. The goal of the proposed effort is to solve the GCD attack RSA flaw. Employing an exclusive OR(XOR), the RSA private and public keys were coupled with the ECC private and public keys, and new hybrid private and public keys were created. A new hybrid private key could be used to decrypt RSA data. The suggested method has the advantage of creating three tiers of cryptographic keys, which can avoid the GCD attack in RSA. When compared to the RSA algorithm's straight forward key creation, the key strength is greater.

Sutikno et al. (1998) discussed the implementation of elliptic curves cryptosystem. After Diffie and Hellman introduced public key cryptography, T. ElGamal introduced a discrete logarithm problem-based public key cryptosystem approach in 1985. Neil Koblitz and Victor Miller proposed elliptic curve cryptosystems for the first time in 1985, on their own. This is because they used elliptic curve groups for arithmetic, elliptic curve cryptosystems are unique. The discrete logarithm issue in a set of points on an elliptic curve formed over a finite field is the basis for this cryptosystem. In an elliptic curve group, the discrete logarithm issue looked to be significantly more difficult than in other groups.

Schroeppel et al. (1995) explained how to build the Diffie-Hellman key exchange algorithm using a set of points on an elliptic curve over the field F_{2n}. A software version of this that used $n = 155$ might be optimized to achieve slightly quicker computation rates with a similar level of security than non-elliptic curve versions. The highly eddicient implementation was made possible by F_{2n} quick calculation of reciprocals.

Kumar et al. (2016) proposed a novel image security algorithm based on elliptic curve cryptography (ECC) and DNA encoding, with the incressed usage of media in communications, image encryption was required to protect against attacks. The RGB image was first encoded using DNA encoding, then asymmetric encryption using elliptic curve Diffie-Hellman encryption was applied (ECDHE). For analysis, the suggested technique was used to standard test photos. Key spacing, key sensitivity, and statistical analysis were all examined. According to the findings, the suggested method can withstand exhaustive attacks and is suitable for real applications.

Bellare et al. (2000) addressed the security of public-key cryptosystems in a "multi-user" scenario, specifically in the presence of assaults utilizing the encryption of related messages using distinct public keys, as demonstrated by Hastad's classical RSA attacks. They demonstrated that security in a single -user context entails security in a multi-user setting if the former is

regarded in the broad sense of "indistinguishability", pinning down a number of methods that are guaranteed to be secure against hastad-type assaults.

ElGamal (1985) offered a novel signature technique, as well as a Diffie-Hellman key distribution scheme implementation that resulted in a public key cryptosystem. The difficulty of computing discrete logarithms over finite fields provided the security for both systems.

Mahto et al. (2016) described data transmission technologies that were transformed, as a result of this. It had provided platform for communicative parties to swiftly communicate information while simultaneously providing an opportunity for adversaries to strike. Cryptographic techniques were used to guarantee secrecy, integrity, and authentication to insecure data at rest, compares and contrasts the security of two widely used public-key cryptography systems, rivest shamir adleman (RSA) and elliptic curve cryptography (ECC). Rivest shamir adleman (RSA) was the first-generation public-key cryptography that was immensely popular since its introduction, while ECC has only lately gained prominence. The integer factorization issue (IFP) was used to secure the RSA cryptosystem, while the elliptic curve discrete logarithm problem (ECDLP) was used to secure the ECC cryptosystem. The major merit of ECC over RSA was because the best-known technique for solving the elliptic curve discrete logarithm problem (ECDLP) requires full exponential time, whereas RSA's (IFP) takes sub-exponential time. This means that ECC can employ far fewer parameters than RSA while maintaining the same level of security. For instance the RSA technique requires a key to produce 112 bits of protection. ECC requires a key size of 224-255 bits, while RSA requires 2048 bits.

Danger et al. (2013) summarized elliptic curve encryption is subject to side-channel attacks. These attacks took use of biases in a variety of leakages, including power consumption, electromagnetic emission, execution time, and so on. To counter known assaults, countermeasures must be integrated. Because no one countermeasure can protect against all types of attacks, many of them must be integrated. However, because each has a hefty price tag, they can not all be used at the same time. Depending on the situation and the trade-off between security and performance, it is crucial to choose countermeasures carefully. The side-channel attacks on elliptic curve cryptography are summarized in this work, along with countermeasures. The cost in terms of time and space is stated for each countermeasure. Some assaults, such as the doubling attack, are clarified; others, such as the horizontal SVA, are enhanced; and new attacks, such as the horizontal attack against unified formulae, are detailed.

Chapter 3

Methodology

Elliptic curve cryptography was discovered in 1985 by Victor Miller and Neil Koblitz as an Elliptic Curve Method (ECM) was applied on cryptography known as Elliptic Curve Cryptography. Cryptography based on Elliptic Curves (ECC) has the unique feature of being the finest up to now.It can be solved using a well-known approach that takes full exponential time to complete.The Elliptic Curve Logarithm is responsible for its safety.In a group specified by the Discrete Logarithm Problem (DLP) over a finite field, using points on an elliptic curve,To attain the same level of security as traditional public key cryptography techniques, the required key size must be drastically reduced.As a result of that,an appropriate public key scheme would be one that is cost-effective in terms of compute and key sizes,Compared to other public key cryptosystems, ECC offers the highest strength-per-bit. The smaller the key, the less bandwidth, memory, and processing power are used.

3.1 Group Law

We first consider the case of Weierstrass elliptic curves. Let L be a line, with the following rules:

1. The identity element is $0 = [0 : 1 : 0]$.
2. If A, B, C are 3 distinct points of $L \cap E$ then $A + B + C = 0$.
3. If $L \cap E$ consists of exactly 2 points A and B, and L is tangent to E at A, then $A+A+B = 0$.
4. If $L \cap E$ is just a single point A and L is tangent to E at A, then we have $A + A + A = 0$. It is commonly known that when the points on an elliptic curve are added together, they form an abelian group. Let E be the Weierstrass equation for an elliptic curve.

3.1.1 Weierstrass Equation

Weierstrass equation is given as;

$$Y^2 = X^3 + aX + b \tag{3.1}$$

The addition rule is given below: For all P, Q, E

(i) $O + P = P$ and $P + O = P$. (So O serves as the identity element.)

(ii) -O = O.

(iii) If P = (x_1, y_1) :f: 0, then -P = $(x_1, -y_1 - a_1x_1 - a_3)$. (Note that P and $-P$ are the only points on E with x-coordinate equal to X_1

(iv) If $Q = -P$, then $P + Q = O$.

(v) If $P \neq O, Q \neq O, Q \neq -P$, then let R be the third point of intersection (counting multiplicities) of either the line $\overline{PQ}$ if $P \neq Q$, or the tangent line to the curve at P if $P = Q$, with the curve (as usual, the tangent line to the curve $f(x, y) = 0$ at $P = (a, b)$ is the line $\frac{\partial f}{\partial x}(P)(x - a) + \frac{\partial f}{\partial y}(P)(y - b)= 0$. Then $P + Q = -R$

3.1.2 Group Properties

Elliptic curves are the set of points that are obtained as a result of solving an elliptic curves function over a predefined space.

Definition:

An elliptic curve over K is a nonsingular projective algebraic curve E of genus 1 over K with a chosen base point o in E.

Some points in elliptic curve:

• Elliptic curves have point with coordinated in any finite field that is rational point, real point, complex point and finite point.

• Elliptic curves with point with point in F_p are finite group.

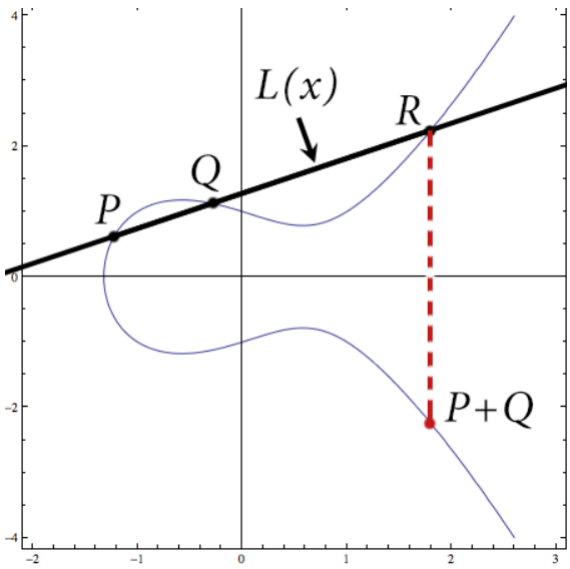

Figure 3.1: Geometric figure of an Elliptic curve

3.2 Using one solution to find various solution to an elliptic curve

The elliptic curve is defined by a certain type of cubic equation with two variables. The set of rational solutions to the equation has an extremely interesting structure, including the group law.

Taking the figure below:

We are to find all the point of on the figure below,and doing so we first find the values of a

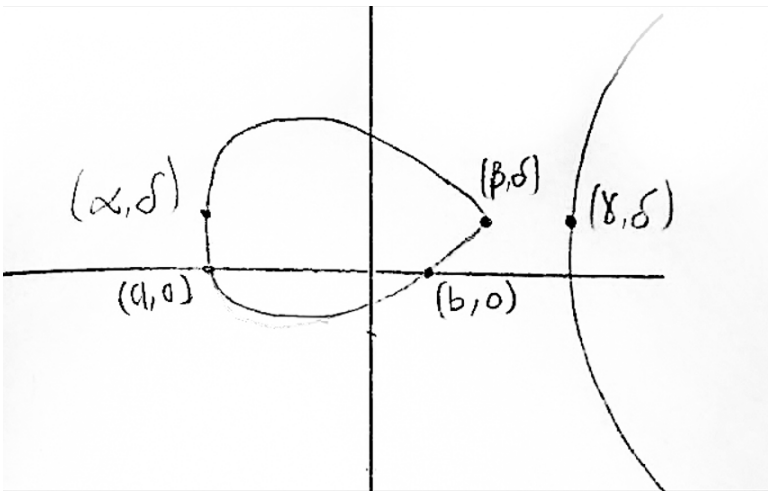

Figure 3.2:

and b

Finding the value of a and b

Given:

Let $Y = 0$

$0^2 - 0 = x^3 - x$

$x(x+1)(x-1) = 0$

From here, $a = -1$ and $b = 1$

Find the values of δ

Adding $\frac{1}{4}$

$Y^2 - Y + \frac{1}{4} = X^3 - X + \frac{1}{4}$

$(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$

From $(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$,

$\delta = \frac{1}{2}$

Why the curve is symmetric about Y=δ

$(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$

$Y = \frac{1}{2} \pm \sqrt{X^3 - X + \frac{1}{4}}$

Note: if $\left(X, \frac{1}{2} + \sqrt{X^3 - X + \frac{1}{4}}\right)$ is on the curve, so is the point $\left(X, \frac{1}{2} - \sqrt{X^3 - X + \frac{1}{4}}\right)$.

Finding a cubic equation with roots α, β, γ

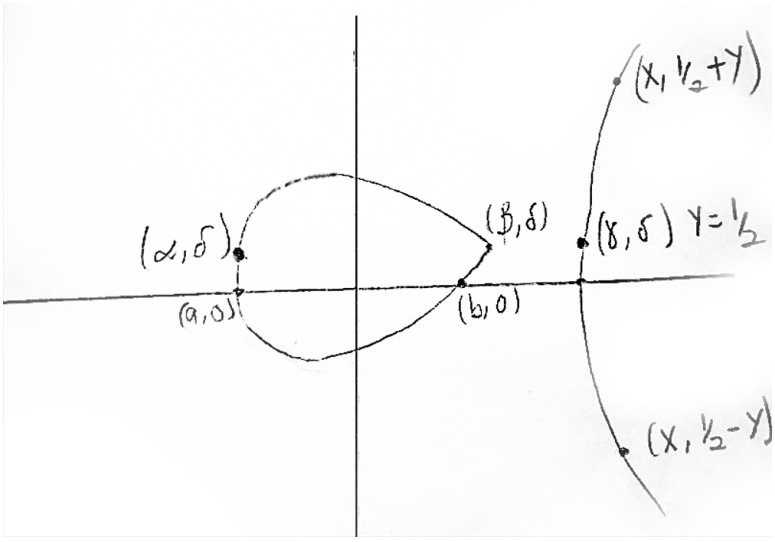

Figure 3.3:

Taking the second line obtained from the object to be our x-axis

Let $Y = \frac{1}{2}$

$\left(\frac{1}{2} - \frac{1}{2}\right)^2 = X^3 - X + \frac{1}{4}$

$X^3 - X + \frac{1}{4} = 0$

Find $\alpha + \beta + \gamma$

$X^3 - X + \frac{1}{4} = (X - \alpha)(X - \beta)(X - \gamma)$

$X^3 - X + \frac{1}{4} = X(X^2 - (\alpha + \beta)X + \alpha\beta(X - \gamma)$

$X^3 - X + \frac{1}{4} = X^3 - (\alpha + \beta + \gamma)X^2 + ...$

$X^3 - X + \frac{1}{4} = -(\alpha + \beta + \gamma) = 0$

$\alpha+\beta+\gamma = 0$

Let C be the circle with (α, δ),(β,δ) as ends of a diameter.

Equation of a circle: $a^2 + b^2 = r^2$

Put the centre $\left(\frac{\alpha+\beta}{2}, \frac{1}{2}\right)$ into the equation of a circle

Centre $\left(\frac{\alpha+\beta}{2}, \frac{1}{2}\right)$

$\left(X - \frac{\alpha+\beta}{2}\right)^2 + \left(Y - \frac{1}{2}\right)^2 = \left(\frac{\beta-\alpha}{2}\right)^2$

Showing that C intersect S at two other points and finding the X-coordinate in terms of δ

$\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$

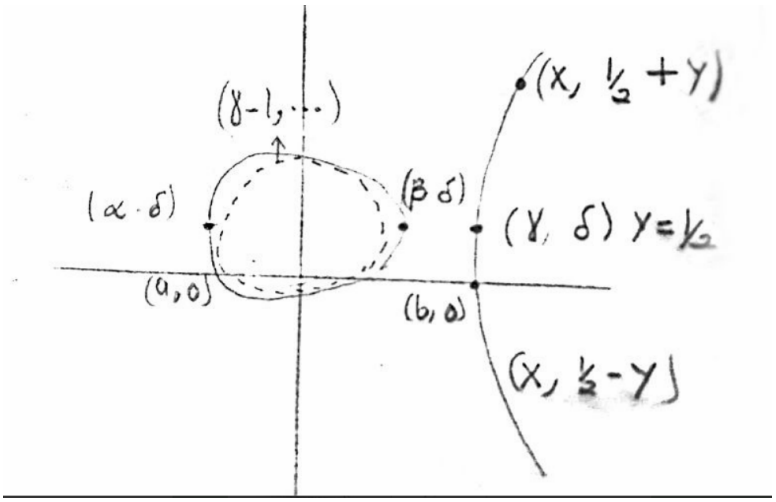

Figure 3.4:

Note we know that α and β are solutions to $\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$,let S be the 3rd solution

Since the coefficient of X^2 in $\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$ is 1

So $\alpha + \beta + S = -1$

$-\gamma + S = -1$

S=γ-1

3.3 Finding point algebraically

For any points P and Q on an elliptic curve, EC group law point addition can be used geometrically to find various points in the plane. If we draw a line through P and Q, this line will intersect the elliptic curve at a third point (-R). The reflection of this point about x-axis, R is the addition of P and Q.

Bezout's Theorem establishes that the intersection of P and Q is valid and it states; Suppose that X and Y are two plane projective curves defined over a field F that do not have a common component (X and Y are defined by polynomials whose greatest common divisor is a constant). Then the total number of intersection points of X and Y with coordinates in an algebraically. The closed field E, which contains F, counted with multiplicity, is equal to the product of the degrees of X and Y. This makes the formation of the third point R after the intersection of P and Q.

Properties

1. If a line intersects two points, it intersects a third.

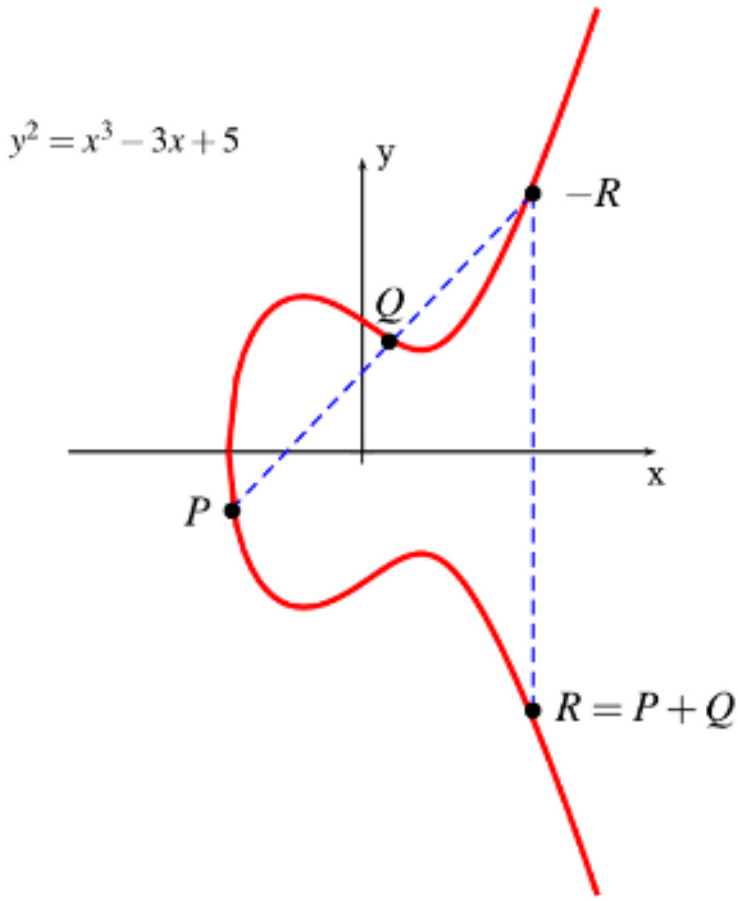

Figure 3.5:

2. If a line is tangent to the curve, it intersects another point.

3. All vertical lines intersect the point at infinity.

4. The curve has an ever-increasing slope after a point of inflection.

The Addition Law

The curve E under the point of addition is an abelian group with these properties:

1. Identity-at infinity

2. Inverse-the reflection of a point.

3. Associativity

4. commutativity

First step in finding the equation is to define the line.

$P = (X_1, Y_1)$ and $Q = (X_2, Y_2)$

• The point-slope from the line.

$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$

$y - y_1 = \lambda\ (x - x_1)$

• The slope-intercept form of the line.

$y = \lambda$x - $\lambda\ x_1 + y_1$

$\beta = y_1$ - $\lambda\ x_1$

$y = \lambda$x+β

Finding x_3 and y_3 of points 3

y=$\lambda\ x + \beta$

squaring both sides

$y^2 = (\lambda$x+$\beta)^2$

The curve $y^2 = x^3 + ax + b$

$(\lambda$x+$\beta)^2 = x^3 + ax + b$

$0 = x^3 - \lambda^2 x^2 - 2\lambda x\beta - \beta^2$+ax+b

x_1,x_2,x_3 *are the roots.*

$(x_1 + x_2 + x_3) = \lambda^2$

$x_3 = \lambda^2 - x_1 - x_2$

$y_3 = \lambda(x_1 - x_3) - y_1$

3.3.1 Identity

Definition:

Let P be the elliptic curve E, and O be the point at infinity,then P+O=O + P Finding the point at infinity 0

1. Adding infinity results in a vertical line.
2. Then find the third point of intersection.
3. After the reflection the result turns to be the original point.

$P + \infty = P$

The figure below illustrate this concept:

3.3.2 Inverse

An operation (+) is said to be invertible within a set if each element P has an element with the same set P^{-1} such that $P + P^1 = O$ where O is the identity of the operation within the set.In

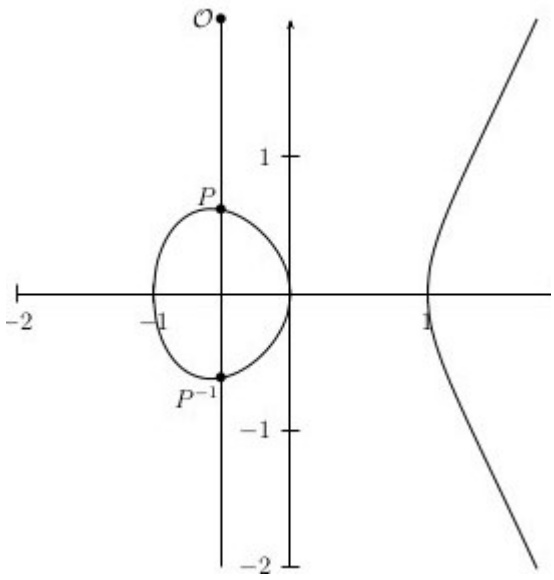

Figure 3.6:

this case,P and P^{-1} are said to be inverses of each other.

$P+-P=\infty$

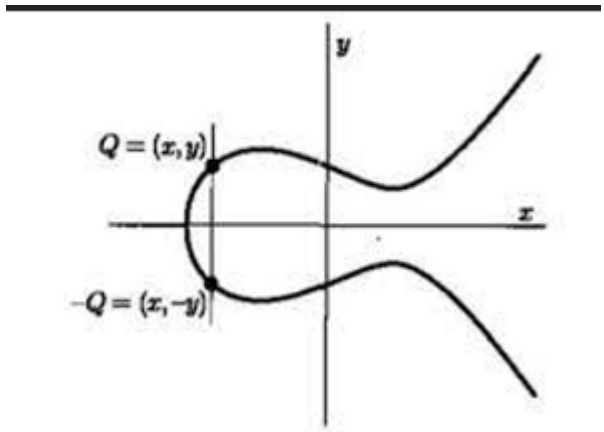

Figure 3.7:

The reflection of a point is it inverse

so, if we define

P=(x,y) and $-P$=(x,$-y$)

$P - P$=∞

3.3.3 Associative

An operation is said to be associative if $P + (Q + R) = R + (P + Q)$ for all P,Q,R within a set.The steps used to find the operation of normal addition (+) on associativity on a geometry

is below:

1. Taking a line through P and Q we get $P+Q$ and it reflection is $P+Q$.
2. Taking a line through P and Q we get $P+Q$ and it reflection is $P+Q$.
3. Taking a line through Q and R we get $Q+R$ and it reflection is $Q+R$.
4. Taking a line through $Q+R$ and P it turns $P+(Q+R)$.
5. Taking a line through $P+Q$ and R it turns $R+(P+Q)$.
6. The line $R+(P+Q)$ and $P+(Q+R)$ intersect at a point whose reflection form a vertical line thus at ∞.
7. This makes $R+(P+Q)$ and $P+(Q+R)$ equal. Taking a line through Q and R we get $Q+R$ and it reflection is $Q+R$.
8. Taking a line through $Q+R$ and P it turns $P+(Q+R)$.
9. Taking a line through $P+Q$ and R it turns $R+(P+Q)$.
10. The line $R+(P+Q)$ and $P+(Q+R)$ intersect at a point whose reflection form a vertical line thus at ∞.
11. This makes $R+(P+Q)$ and $P+(Q+R)$ equal.

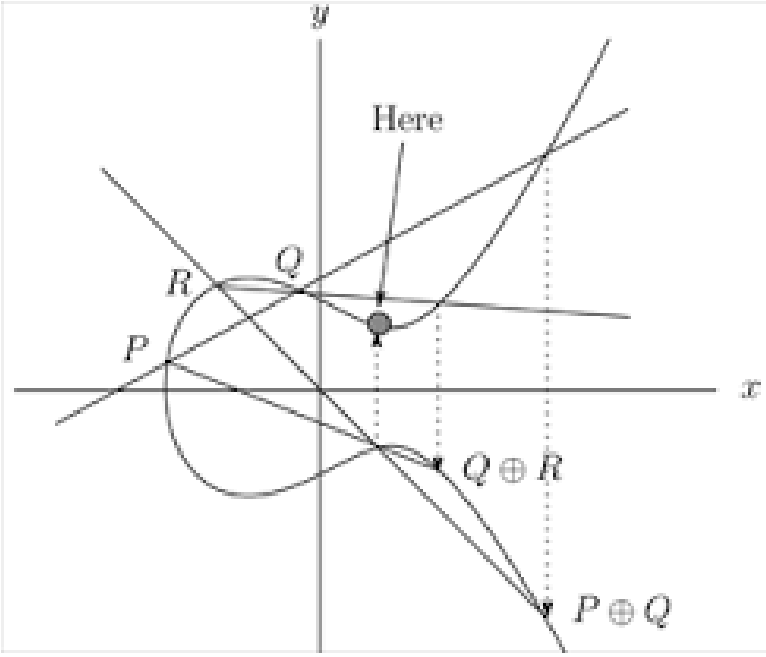

Figure 3.8:

3.3.4 Point Doubling

When the points P and Q are of the same coordinates, addition is identical, except there is no well-defined straight line between P and Q, so the operation is closed using the limiting case, the tangent to the curve, E, at P and Q.

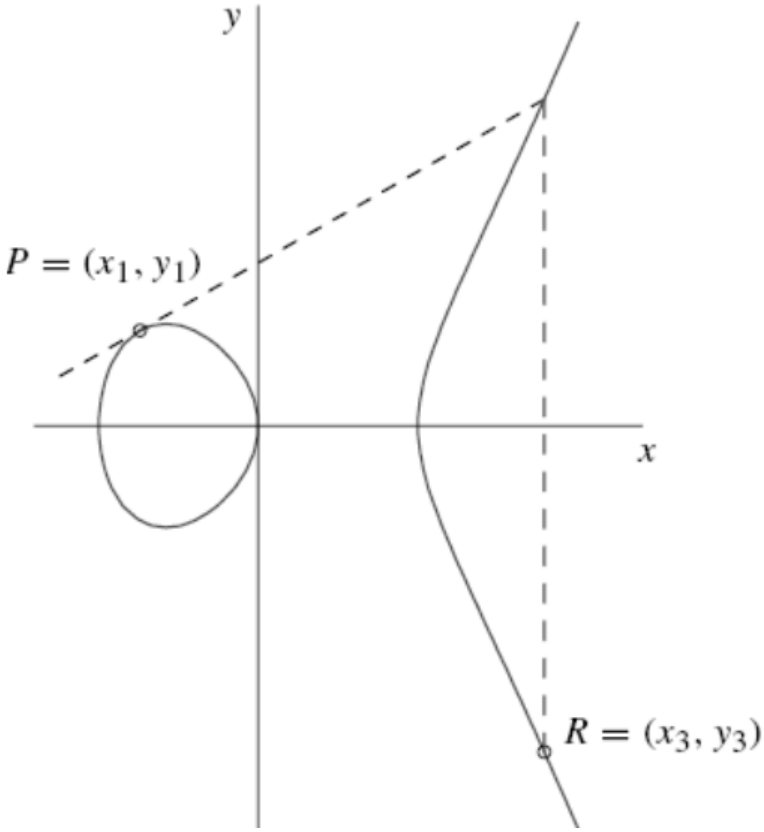

Figure 3.9:

Point doubling is calculated using following equation:

$$2P = (x_3, y_3)$$

,where

$$x_3 = \lambda^2 - 2x_1$$

$$y_3 = \lambda(x_1 - x_3) - y_1$$

$$\lambda = \frac{3x_1^2 + a}{2y_1}$$

Thus:if $P = Q$

The properties of the group law all works together for good for elliptic curve cryptography.It basically works good under the addition law and point doubling,in elliptic curve cryptography.This makes the security of elliptic curve cryptography more reliable because a series of generator points has to be traced back to the original point, and it will be very difficult to do so based on the repeated point addition and point doubling.

3.4 cyclic group

Let G be a finite group of order n is called cyclic if there is an identity element $a \in G$ such that the cyclic subgroup generated by a is the entire group G. In other words, $G = a^n : n \in Z$. Such an element a is called a generator of G.

Note: A cyclic group is abelian but a group that is abelian is not necessarily.

To do some work with the points of elliptic curves, you will need to know what an elliptic curve is and what a group law is. The cyclic subgroup, the base point, and the order can be determined using the points of an elliptic curve when they are restricted to finite fields of integers modulo a prime.

3.4.1 The Key Pair Generation of ECC

Elliptic curve cryptography requires the following parameters below:

1. The finite field's size is specified by the prime p.
2. The elliptic curve equation has coefficients a and b.
3. The cyclic subgroup is generated by the generator G.
4. n is of the subgroup's order.

The private key is a random integer d selected from the range $1, ..., n-1$, where n is the subgroup order. The public key is $PK = dG$, where G is the subgroup's generator. Finding PK is simple if we have d and G, as well as the other domain parameters. Finding the private key d is difficult if we know PK and G since it requires solving the discrete logarithm issue.

3.5 Diffie - Hellman Key Exchange on elliptic curve cryptography

Suppose Budu-Biney and his Bank establish a common key with his bank with the aim to exchange data as part of symmetric encryption scheme.

1. Budu-Biney and his Bank agree to use a specific curve (elliptic curve E) over a finite field Fq that makes the discrete logarithm problem impossible in E (F_q). They all agreed on a point $P \in (F_q)$ indicating that the P-generated subgroup has a big order.
2. Alice selects a secret $b \in Z$ and enumerate bP = Pb and sends Pb to his bank.

3. Bank select a secret $c \in Z$ and enumerate $cP = Pc$ and sends Pc to Budu-Biney

4. Budu-Biney enumerate $bPc = bcP$.

5. The Bank enumerate $cPb = bcP$. To extract a key from bcP, Budu-Biney and his bank utilize a widely accepted approach, such as the final 256 bits of $bcP's$ x-coordinate.

On the elliptic curve E, the finite field F_q and the points P, bP, and cP.The only way an intruder can get Budu-Biney's secret key is to use P, bP, and cP to determine bcP. If the intruder can find the discrete logs over E(F), them the intruder could use P and bP to find Budu-Biney's secret key b.Then the third person could now find $b(cP)$ to get bcP. Solving discrete logs knowledge is required to solve it and without it is assumed to be infeasible to fine over E(F_q).

To determine whether $Q = bcP$ given P, bP,$cP \in E(F_q)$ and a point $Q \in E(F_q)$ in a comparable problem known as the Decision Diffie-Hellman problem. In other words, can you verify that bcP is right. In some situations, as shown below, the Weil pairing can be utilized to solve the Decision Diffie-Hellman issue.

3.6 ElGamal Public Key Encryption

Assume Budu-Biney wants to communicate with his Bank.To begin, the bank creates it public key in the manner described below.The bank selects an elliptic curve E over Fq that makes the discrete logarithm issue for E(Fq) impossible, as well as point $P \in E(Fq)$ P have a large prime order.The bank also selects a secret $a \in Z$ and calculates $B = aP$. As a result, Budu-Biney's public and private keys are E, Fq, P, and B, respectively.

To send a message to the bank, Budu-Biney follows the steps below.

1. The Bank's public key is downloaded.

2. Use a point $M \in E$ to convey her meaning (Fq).

3. He selects a secret $k \in Z$ and calculates $M_1 = kP$.

4. $M_2 = M + kB$ is calculated.

5. $M_1 and M_2$ are sent to his bank.

The bank calculates the cipher text in order to decrypt it. $M_2 - sM_1 = (M + kB) - s(kP) = (M + skP) - skP = M$

if the intruder has the bank's public key as well as the points $M_1 and M_2$, but there appears to be no way for the intruder to access M if the intruder does not know how to solve discrete

logs. Budu-Biney must use a different $k \in Z$ each time he sends a message to the bank, which is an essential note about the $k \in Z$ he chooses at random.The intruder can quickly detect Budu-Biney's usage of the same integer k if he computes M and M0 with the same value of k. The bank may then use $M_2 M_{20} = M_0$ to recover the message M by remembering that $M_0 = MM_2 + M_2$

3.7 ElGamal Digital Signatures

Let's say Budu-Biney wants to sign a bank document electronically. One method is to digitalize Budu-Biney's signature. However, the bank can copy Budu-Biney's digital signature and then attach it to any document. The steps that are required to be used so that such a digital signature can only be used once. However, someone should be able to verify that the signature is genuine and that Budu-Biney is the one who signed the document. One solution to the problem is based on the discrete logarithm problem's difficulty.Budu-Biney first picks an elliptic curve E over a finite field F_q in which the discrete logarithm issue is impossible to solve (Fq).He also chooses $A \in E(F_q)$ as a location where $|A| = N$ is a huge prime.Budu-Biney also selects $a \in Z$ secret and calculates the location $B = aA$. Finally,He selects a function $f : E(F_q) \longrightarrow Z$ that has the property of having a huge picture and just a small number of inputs that result in a certain output.As a result, Budu-Biney's public key is composed of the letters $E, F_q, f, A, andB$. The secret $a \in Z$ is his private key. The order of point A,On the other hand, it does not have to be made public. As a result, Budu-Biney does the following to sign the bank document:

1. The bank document is represented as $m \in Z$. If $m > N$, Budu-Biney will select a larger curve.

2. R = kA is solved by picking a random $k \in Z$ and setting $gcd(k, N) = 1$.

3. Solves $s \equiv k^{-1}(m - af(R))(modN)$

After the message has been signed (m, R, s). It is worth noting that Budu-Biney is not interested in keeping the message m a secret. If he wants to do so, he will need to employ some type of encryption.The bank then confirms that the communication is from Budu-Biney by saying:

1. The bank downloads Budu-Biney's information

2. Calculates $V_1 = f(R)B + sRandV_2 = mA$.

3. If $V_1 = V_2$, then the bank declares that the signature is authentic.

R is the same in both signatures, thus you will have to do it twice. The following are the equations for s and $s^{'}$:

$$ks \equiv (m - af(R))(modN)$$

$$ks^{'} \equiv (m^{'} - af(R))(modN)$$

Taking the difference between these two match gives $k(s - s^{'}) \equiv (m - m^{'})(modN)$.Set $d = gcd(s - s^{'}, N)$ then there are d possible values of k.The intruder tries all of them until he finds R = kA. He can then locate an as previously once he knows k.

Chapter 4

According to Silverman and Tate (2015),to find an integer $c \in Z$ and search for Diophantine equation solutions.y^2 - x^3= c.Suppose we are looking for a solution in the rational numbers $x, y \in Q$.If (x, y) is a solution with rational x and y and $y \neq 0$, it is simple to verify that the pair is correct.

$$(\frac{x^{4-8cx}}{4y^2}, \frac{-x^6 - 20cx^3 + 8c^2}{8y^3})$$

is an answer to the identical equation in rational numbers.If the original solution is $x, y \neq 0$, then repeating the process will provide an unlimited number of different solutions.if an integer can be written using nonzero rational numbers as the difference between a square and a cube

4.1 Application of rational numbers on elliptic curves in cryptography.

A non-singular projective curve of genus one with a determined base point is called an elliptic curve. Put a field k, such as the rational numbers Q, a finite field F_q, or even a function field C(t), in fancier terminology for the inspired. An elliptic curve E defined over k is defined as the functor that connects fields K containing k to an algebraic set. This is Because we can make a linear change of variables, an elliptic curve is represented by a cubic equation of the form $y^2 = x^3 + Ax + B$ for some $A, B \in k$ fulfilling $4A^3 + 27B^2 \neq 0$ if k is not of characteristic 2 or 3. Indeed, on the curve with z = 0, $O = (0 : 1 : 0)$ is the only projective point $P = (x : y : z)$.

4.1.1 Fermat curves

First of all, suppose the curve $a^3 + b^3 = c^3$ over a field $k = Q$, By substitution $a = 32 - 8y$, $b = 40+8y$, and $c = 24x$, determine the elliptic curve $y^2+y = X^3-7$. The curve $3a^3+4b^3 = 5c^3$ on the other side, is not an elliptic curve over Q because it has no Q rational solutions other than the degenerate one, namely $a = b = c = 0$. However, this equation does define a genus one projective curve. For various exponents n, the curve $a^n + b^n = c^n$ over Q has piqued people's interest. When n is a large enough integer, Pierre de Fermat question if there were any

nonzero Q rational answers. When $n = 2$, any Pythagorean Triple, such as $(a, b, c) = (3, 4, 5)$, will suffice as a desirable answer. We have seen that when $n = 3$, elliptic curve arise, but this isn't the only exponent where this occurs. Surprisingly, every solution (a, b, c) provides a point $P = (x, y, 1)$ on the elliptic curve E when $n = 7$. So $E : y^2 + xy = x^3 - x^2 - 107x + 552$ by a hard and nontrival-substitution

$$x = 49\frac{(a^2 + b^2 + c^2 + ab + ac + bc)^2 + abc(a + b + c)}{(a + b + c)^4} - 12$$

,

$$y = \frac{7(x + 12)(a^2 + b^2 + c^2) - x(a + b + c)^2}{(a + b + c)^2}$$

By focusing on Q-rational P on E, one can one can identify all Q-rational solutions to $a^7 + b^7 = c^7$.

4.1.2 The Mordell Weil group

The structure of the abelian group $E(K)$ is an interesting subject to study. For finite extensions K of Q, Louis Mordell's famous theorem states that $E(K)$ is finitely created, that is, $E(K)torsZr$ for some finite group $E(K)tors$ consisting of torsion elements and some non-negative integer r called the rank.

$$y^2 = x(x + d_1)(x + d_2)$$

Mazur's Theorem asserts that E(Q)tors (Z/2 Z) (Z/2 m Z) for some m = 1, 2, 3, or 4 for some m = 1, 2, 3, or 4. We can choose $T_1 = (0 : 0 : 1)$ as a generator of order 2, leaving only T_2 as a generator of order 2 m to discover.The rank is much more difficult to determine: there is an injective group homomorphism, which is based on a Mordell notion.

4.2 The Discrete Logarithm Problems

There are a few unsolved issues in the field of discrete logarithms in cyclic groups. The issues arose after the Diffie-Hellman protocol was introduced in 1976. These unsolved issues are listed below.

1. The Problem of Discrete Logarithms. Let G = $g^r : 0 \leq r \leq n$ denote an n-element random cyclic group created by $g > 1$. Find the integer r = log(s) from a primitive element g and a random element $s = g^r G$.

2. The DH Problem in Computation Find g^{rs} for n, g, and a random pair of elements g^r

and g^s if you have $nandg$.

3. DH's choice Problem. Determine whether $g^t = g^{rs}$ if you have n, g, and a random triple of components g^r, g^s, and g^t.

4. Inappreciable Assumption. In polynomial time, a randomly generated triple (g^r, g^s, g^{rs}) $\in GxGxG$ is indistinguishable from a randomly picked triple (x, y, z) $GxGxG$.

Computing the discrete logarithm on a random cyclic group and other tough problems are examples of intractable problems. When $\in > 0$, the fastest algorithms run in ($n^{1/2} + \in$) exponential time. The running time complexity is the number of group operations required to calculate a single discrete logarithm instance. Over the years, a lot of research has been done on this subject, but none of it has reduced the temporal complexity of the discrete logarithm in random cyclic groups. [book()].

4.3 Attacks on Discrete logs

It is easy to show that a number of these attacks are analogous to computing discrete logarithms over GF_p. Computing discrete logarithms or breaking the distribution scheme are not yet comparable to breaking the signature scheme. However, none of the assaults described in this section appear to have succeeded in cracking the system. There will be two types of attacks. In the first group, we show several methods for recovering the secret key x, and in the second group, we show some approaches for forging signatures without recovering x.

4.3.1 Attacks Aiming to Recover x

Attack 1: Given $m_i i = 1, 2, ..., l$ information compiled with the corresponding signatures $(r_i, s_i) : i = 1, 2, ..., l$ if an intruder tries to compute l equation of the form $m \equiv xr + ksmod(p-1)$.There are $l+1$ unknowns, and not all of the possible solutions to the system of equations have been found. Each value for x produces a system of linear equations with a diagonal matrix of coefficients, and hence each value for x provides a solution for the k_i.Recollecting $xmodq$ requires an exponential number of message-signature pairs if $p-1$ has at least one large prime factor q. Note:If any k is used twice in signing, the system of equations can be solved once and x retrieved. To ensure the security of the system, no value of k should be used twice.

Attack 2: Trying to compute equations of the form $a^m \equiv y^r r^s modp$ always the sames as solving discrete logarithms over $GF(p)$, when both unknowns appear in the exponent.

Attack 3: Then the intruder will try to figure out and develop some linear dependencies in the unknown variables $k_i, i = 1, 2, ..., l$.This is the same as solving discrete logarithms

since if $k_i \equiv ck_j mod(p-1)$ hence $r_i \equiv r_j^c modp$.if c can be solved then solving discrete logarithms is easy.

4.3.2 Attacks for Forging Signatures

In order to satisfy a given document m, a forger could try to obtain r, s. If $r = a^j modp$ is constant for some j selected at random, computing s is comparable to solving a discrete logarithm problem over GF_p. The equation can be used to compute r if the forger fixes s first.

$r^s y^r = Amodp$.

Attack 4: Although it has not yet been proven that solving equation $r^s y^r = Amodp$ for r is at least as difficult as computing discrete logarithms, we believe that solving $r^s y^r = Amodp$ in polynomial time is not possible. The reader is challenged to come up with a polynomial time algorithm for addressing the problem $r^s y^r = Amodp$.

Attack 5: We haven't been able to find an efficient solution for both r and s at the same time.

Attack 6: The signature mechanism allows for the following attack: if an attacker knows one authentic signature for one message, he or she can create other real signatures and messages. The system is not broken because this strategy prevents an intruder from signing any message. This property can be avoided by giving m a specified structure or applying a one-way function to the message m before signing it. The system is not broken because this strategy prevents an intruder from signing any message. This property can be avoided by giving m a specified structure or applying a one-way function to the message m before signing it.

Given a signature for a message (m),then

$$a^m = y^r r^s modp$$

Taking an integer A,B and C arbitrarily such that (Ar -Cs) is relatively prime to p - 1.then set

$$r^{'} \equiv r^A a^B y^C modp,$$

$$s^{'} \equiv sr^{'}/(Ar - Cs)mod(p - l)$$

,

$$m^{'} \equiv r^{'}(Am + Bs)/(Ar - Cs)mod(p - 1)$$

. since it is valid that $(r^{'}, s')$ signs the message $(m^{'})$.

solve

$$y^{r'}r'^{s'} \equiv y^{r'}\left(r^A\alpha^B y^C\right)^{sr'/(Ar-Cs)}$$

$$y^{r'}r'^{s'} \equiv \left(y^{r'Ar-r'Cs+r'Cs}r^{Asr'}\alpha^{Bsr'}\right)^{1/(Ar-Cs)}$$

$$y^{r'}r'^{s'} \equiv \left((y^r r^s)^{Ar'}\alpha^{Bsr'}\right)^{1/(Ar-Cs)}$$

$$y^{r'}r'^{s'} \equiv \alpha^{m'}$$

$$r' \equiv \alpha^B y^C \bmod p,$$

$$s' \equiv -r'/C \bmod (p-1),$$

$$m' \equiv -r'B/C \bmod (p-1).$$

It can be shown that (r', s') signs (m').

4.3.3 Baby Step, Giant Step

Shanks pioneered the first discrete logarithm approach, which requires around $\sqrt{n}$ steps and $\sqrt{n}$ storage to perform well for fairly large n. Shanks' algorithm, now known as the Baby Step, Giant Step algorithm, is as follows:

1. Put an integer $m \geq \sqrt{n}$ and evaluate mP.
2. create and save a list of i_P for $0 \leq i \leq m$
3. Find the point $Q = jmP$ for $j = 0, 1..., m-1$ till one corresponds with a point from the save list.
4. If $i_P = Q - jmp$, then $Q = kP$ with $k = (i + jm)(modn)$.Hence $Q = i_P + jmP = (i + jm)P = kp$.

If we put an integer $m\sqrt[1]{n}$,then $m^2\sqrt[1]{n}$ this algorithm will work suppose that the solution k is such that $0 \leq k < m^2$.Taking $k = k_0 + mk$ where $k_0 \equiv k(modm)$ and $k_1 = (k - k_0)/m$.when $i = k_0 andj = k_1$ respectively then $0 \leq k_1 < m$

We get

$$Q - jmp = Q - k_1mp = kP - k_1mP = (k - k_1m)P = k_0P$$

then the point corresponds with another point.

Assuming we only have an upper bound on $n = |G|$, we may use Hasse's theorem to apply this approach to an elliptic curve over Fq.

Example. Let $G = E(F_{41})$, where E is given by $y^2 = x^3 + 2x + 1$. Let $P = (0, 1)$ and $Q = (30, 40)$. By Hasse's theorem,

suppose

$$|41 + 1 - E(F_{41})| \leq 2\sqrt{41} \implies 29.194 \leq E(F_{41}) \leq 54.806$$

, if $n = |G| \leq 54$. then we let $m = 8$ since $m \geq \sqrt{54}$. Points P_i for $1 \leq i \leq 7$ are

$$(0, 1), (1, 39), (8, 23), (38, 38), (23, 23), (20, 28), (26, 9).$$

When $Q = jmpforj = 0, 1, 2$ we evaluate to obtain

$$(30, 40), (9, 25), (26, 9),$$

Where we get a corresponding point to another point, since $j = 2$ corresponds with 7P, we then get

$$Q = (30, 40) = 7P + 16P = 23P$$

.

4.3.4 Pollard's ρ Method

The Baby Step, Giant Step algorithm has a significant disadvantage in that it requires a lot of storage space. Pollard's method takes roughly the same length of time as Baby Step, Giant Step, but requires significantly less storage. Assume that G is a finite group with $|G| = n$ and $|G| = n$. Choose an erroneous function $f : G \implies G$. We start with P_0 and compute iterations from there. $P_{i+1} = f(P_i)$. There will be some indices i_0 j_0 such that $P_{i0} = P_{j0}$ since $|G| = n$. Then,

$$P_{i0+1} = f(P_{i0}) = f(P_{j0}) = P_{j0+1}$$

For every $\rho \geq 0$ $P_{i0+\rho} = P_{j0+\rho}$. As a result, $P_{i i \in N}$ has a periodic period of $j_0 - i_0$ or is a divisor of $j_0 - i_0$. Because the procedure's graphic resembles a Greek letter, Pollard's method is called that. If f is a randomly chosen random function, we should expect to find a match with j0 at most a constant times p. Then we should find a match with j_0 at most a constant times $\sqrt{n}$.

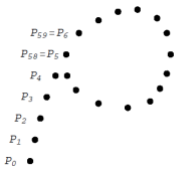

Figure 4.1: Pollard's ρ Method

One technique implementation stores all the points P_i until a match is found. This takes about $\sqrt{n}$ of storage, which is similar to Baby Step, Giant Step. It is feasible to do significantly better with a little extra computation. The idea is that once two indices that differ by d match, all subsequent indices that differ by d will also match. Thus, for $i = 1, 2, 3, ...$ we can compute pairs (P_i, P_{2i}), but we only keep the current pair. The prior pairs will not be kept.

$$P_{i+1} = f(P_i) and P_{2(i+1)} = f(f(P_{2i}))$$

If $i > i_0$ and i is a multiple of d, then 2_i and i are a multiple of d apart. As a result, $P_i = P_{2i}$ is a match. There is a match for $i < j_0$ since $d \leq j_0$. As a result, the number of steps required to discover a match should be at most a constant multiple of $\sqrt{n}$. The issue is determining which function f is appropriate. Divide G into s distinct subsets $S_1, S_2, ...S_s$ of roughly equal size. 2_s randomly generated integers $a_i, b_i (mod n)$.

$$M_i = a_i P + b_i Q$$

define

$$f(g) = g + M_i$$

if $g \in S_i$ Choose two random integers, a_0 and b_0, and set the starting point at $P_0 = a_0 P + b_0 Q$. We keep track of how these points are expressed in terms of P and Q when computing P_j. If

$$P_j = u_j P + v_j Q and P_{i+1} = P_j + M_i$$

, then

$$P_{j+1} = u_j P + v_j Q + a_i P + b_i Q = (u_j + a_i) P + (v_j + b_i) Q$$

so

$$(u_{j+1}, v_{j+1}) = (u_j + v_j) + (a_i, b_i).$$

finding a match $P_{i0} = P_{j0}$, then

$$u_{j0}P + v_{j0}Q = u_{i0}P + v_{i0}Q \implies (u_{i0} - u_{j0})P = (v_{j0} - v_{i0})Q$$

so $gcd(v_{j0} - v_{i0}, n) = d$, then

$$k \equiv (v_{j0} - v_{i0})^{-1}(u_{i0} - u_{j0})(modn/d).$$

This results in d options for k. Because d is usually small, we can test all alternatives until $Q = kP$. Example Let G = E(F1093) where E is the elliptic curve $y^2 = x^3 + x + 1$. using $s = 3$. Let $P = (0, 1)$ and $Q = (413, 959)$. It can be shown that $|P| = 1067$ and we want to find $k \in Z$ such that $Q = kP$. Let $P_0 = 3P + 5Q$, $M_0 = 4P + 3Q$, $M_1 = 9P + 17Q$, and $M_2 = 19P + 6Q$. Let $f : E(F_{1093}) \implies E(F_{1093})$ be defined by

$$f(x, y) = (x, y) + M_i$$

if $x \equiv i(mod3)$.

In this case, x is an integer with the value $0 \leq x \leq 1093$. It is also possible to define $f(\infty) = \infty$. If we find $P_0, P_1 = f(P_0, P_2 = f(P_1)$ we then get:

$$P_0 = (326, 69), P_1 = (727, 589), P_2(560, 365), P_3(1070, 260)$$

,

$$P_4 = (473, 903), P_5 = (1006, 951), P_6 = (523, 938), ...$$

$$P_{57} = (895, 337), P_{58} = (1006, 951), P_{59} = (523, 938), ...$$

S As a result, $P_5 = P_{58}$, and we keep track of the P and Q coefficients we discover. $P_5 = 88P + 46Q$ and $P_{58} = 685P + 620Q$. Then

$$P_{58} - P_5 = 567P + 574Q = \infty$$

$$Hence|P| = 1067$$

We solve

$$-574^{-1}597 \equiv (mod1067)$$

Hence, Q=499P so that k=499

.

4.4 Elliptic Curve Domain Parameters

There are two types of domain parameters for elliptic curves: those over F_p and those over F_{2m}.Elliptic curve domain parameters over F_p and elliptic curve domain parameters over F_{2m} are detailed in Brown (2009) .Elliptic curve domain parameters can be verified as random, i.e., they are obtained in part from the output function applied to a seed value S. Despite the fact that taking random elliptic domain parameters are preferable, other solutions may be used for a variety of reasons, including enhanced performance.

4.4.1 Elliptic Curve Domain Parameters over Fp

The domain parameters of an elliptic curve on a finite field, say Fp, are a sextuple:

$$T = (p, a, b, G, n, h)$$

Taking an integer p and setting out the finite field F_p, two point $a, b \in F_p$ so setting out an elliptic curve $E(F_p)$ is defined by:

$$E : y^2 \equiv x^3 + ax + b(modp)$$

Considering a base point $G = (xG, yG)$ on $E(F_p)$, a certain prime called n which is the order of G, and an integer h that is the cofactor $h = E(F_p)/n$.

4.4.2 Recommended Parameters secp192k1

The sextuple $T = (p, a, b, G, n, h)$ provides the domain parameters of an elliptic curve over Fp for a Koblitz curve secp192k1, where the finite field Fp is specified by:

$$p = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFEE37$$

$$= 2^{192} - 2^{32} - 2^{12} - 2^8 - 2^7 - 2^6 - 2^3 - 1$$

The curve E: $y^2 = x^3 + ax + b$ over F_p is defined by:

$$a = 000000000000000000000000000000000000000000000000$$

$$b = 0000000000000000000000000000000000000000000000003$$

We have the base point G in a compressed form is:

$$G = 03DB4FF10EC057E9AE26B07D0280B7F4341DA5D1B1EAE06C7D$$

the uncompressed form is:

$$G = 04DB4FF10EC057E9AE26B07D0280B7F4341DA5D1B1EAE06C7D9B2F2F6D9C5628A7844163D01$$

the order n of G and it cofactor are:

$$n = FFFFFFFFFFFFFFFFFFFFFFFE26F2FC170F69466A74DEFD8D$$

$$h = 01$$

p is the prime, a and b are the curve's parameters, G is the generator point, n is the number of points in the field, and h is the cofactor for all elliptic curve cryptography.

using weierstrass equation :

$$y^2 = x^3 + ax + b$$

To locate an elliptic curve's private key. The diagram below illustrates how the private key can be used in conjunction with elliptic cure and algorithm knowledge..

Using Python:

The result to the figure above is shown below:

Spyder (Python 3.7)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\hp\Documents

Editor - C:\Users\hp\Documents\king.py

king.py

```python
# secp256k1 domain parameters
Pcurve = 2**256 - 2**32 - 2**9 - 2**8 - 2**7 - 2**6 - 2**4 -1
Acurve = 0;
Bcurve = 7;
Gx = 0x79BE667EF9DCBBAC55A06295CE870B07029BFCDB2DCE28D959F2815B16F81798
Gy = 0x483ADA7726A3C4655DA4FBFC0E1108A8FD17B448A68554199C47D08FFB10D4BB
GPoint = (int(Gx),int(Gy)) # This is our generator point
N=0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141 # Number of points in the field
#Replace with any private key
privKey = 0xA0DC65FFCA799873CBEA0AC274015B9526505DAAAED385155425F7337704883E #replace with any private key

def modinv(a,n=Pcurve): #Extended Euclidean Algorithm/'division' in elliptic curves
    lm, hm = 1,0
    low, high = a%n,n
    while low > 1:
        ratio = high/low
        nm, new = hm-lm*ratio, high-low*ratio
        lm, low, hm, high = nm, new, lm, low
    return lm % n
def ECadd(a,b): # Not true addition, invented for EC. Could have been called anything.
    LamAdd = ((b[1]-a[1]) * modinv(b[0]-a[0],Pcurve)) % Pcurve
    x = (LamAdd*LamAdd-a[0]-b[0]) % Pcurve
    y = (LamAdd*(a[0]-x)-a[1]) % Pcurve
    return (x,y)
def ECdouble(a): # This is called point doubling, also invented for EC.
    Lam = ((3*a[0]*a[0]+Acurve) * modinv((2*a[1]),Pcurve)) % Pcurve
    x = (Lam*Lam-2*a[0]) % Pcurve
    y = (Lam*(a[0]-x)-a[1]) % Pcurve
    return (x,y)
def EccMultiply(GenPoint,ScalarHex): #Double & add. Not true multiplication
    if ScalarHex == 0 or ScalarHex >= N: raise Exception("Invalid Scalar/Private Key")
    ScalarBin = str(bin(ScalarHex))[2:]
    Q=GenPoint
    for i in range (1, len(ScalarBin)): # This is invented EC multiplication.
        Q=ECdouble(Q);
        if ScalarBin[i] == "1":
            Q=ECadd(Q,GenPoint);
    return (Q)
print; print ("************ Private key Generation **********");
print
PublicKey = EccMultiply(GPoint,privKey)
print ("the private key :");
print (hex(privKey));
print;
```

Figure 4.2:

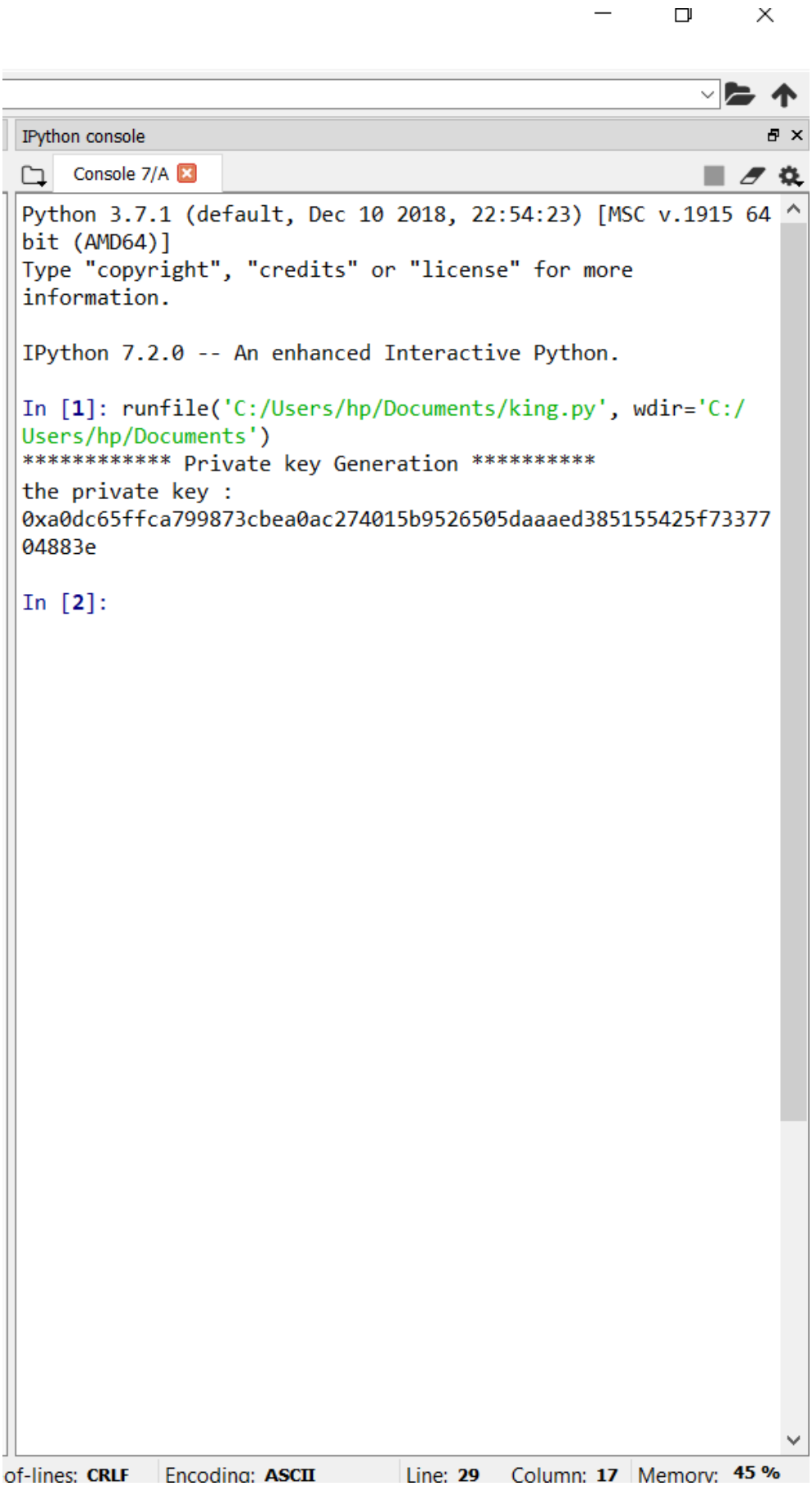
IPython console
Console 7/A
Python 3.7.1 (default, Dec 10 2018, 22:54:23) [MSC v.1915 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.
IPython 7.2.0 -- An enhanced Interactive Python.
In [1]: runfile('C:/Users/hp/Documents/king.py', wdir='C:/Users/hp/Documents')
********** Private key Generation **********
the private key :
0xa0dc65ffca799873cbea0ac274015b9526505daaaed385155425f7337704883e
In [2]:
of-lines: CRLF
Encoding: ASCII
Line: 29
Column: 17
Memory: 45 %

Figure 4.3:

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

Victor Miller and Neal Koblitz independently conceived the notion of utilizing elliptic curves as the basis of a group for the discrete logarithm problem in 1985, and this was the beginning of elliptic curve cryptography.The fact that no sub exponential technique existed to solve the discrete logarithm issue on a suitably designed elliptic curve is the major reason behind Elliptic curve cryptography's appeal. This meant that Elliptic curve cryptography used substantially lower parameters while maintaining the same level of security.Elliptic Curves provide security in the same way as traditional systems (such as RSA) do, but with fewer bits.

The use of elliptic curves in cryptography necessitated a smaller chip size, lower power consumption, and increased speed. In the contest of generating keys, the cyclic group, key pair generation together with the methods of application of rational numbers on elliptic curves in cryptography and finding points algebraically were used.

We also discussed about Diffie - Hellman Key Exchange on elliptic curve cryptography, ElGammal Public Key Encryption and ElGammal Digital Signatures which were geared towards the secure transfer of information. It was also found that, in channeling attacks and security twist the methods used were based on Attacks on Discrete logs, Baby Step, Giant Step, Pollard's p Method. A script was derived for generating private key cryptography.

5.2 Recommendation

Elliptic curve cryptography is considered to be a tool for mathematical studies, specifically in getting to know private key cryptography in ECC to keep confidential information between two parties, along with channel attacks and twist-security attacks based on the details and illustrative codes in this research.

REFERENCES

Afreen, R. and Mehrotra, S. C. (2011). A review on elliptic curve cryptography for embedded systems. *arXiv preprint arXiv:1107.3631*.

Barbulescu, R., Gaudry, P., Guillevic, A., and Morain, F. (2015). Improving nfs for the discrete logarithm problem in non-prime finite fields. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 129–155. Springer.

Bellare, M., Boldyreva, A., and Micali, S. (2000). Public-key encryption in a multi-user setting: Security proofs and improvements. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 259–274. Springer.

Brown, D. (2009). Standards for efficient cryptography, sec 1: elliptic curve cryptography. *Released Standard Version*, 1.

Chen, Z. (2008). Elliptic curve analogue of legendre sequences. *Monatshefte für Mathematik*, 154(1):1–10.

Danger, J.-L., Guilley, S., Hoogvorst, P., Murdica, C., and Naccache, D. (2013). A synthesis of side-channel attacks on elliptic curve cryptography in smart-cards. *Journal of Cryptographic Engineering*, 3(4):241–265.

ElGamal, T. (1985). A public key cryptosystem and a signature scheme based on discrete logarithms. *IEEE transactions on information theory*, 31(4):469–472.

Koblitz, N. (1987). Elliptic curve cryptosystems. *Mathematics of computation*, 48(177):203–209.

Koblitz, N., Menezes, A., and Vanstone, S. (2000). The state of elliptic curve cryptography. *Designs, codes and cryptography*, 19(2):173–193.

Kumar, M., Iqbal, A., and Kumar, P. (2016). A new rgb image encryption algorithm based on dna encoding and elliptic curve diffie–hellman cryptography. *Signal Processing*, 125:187–202.

Mahto, D., Khan, D. A., and Yadav, D. K. (2016). Security analysis of elliptic curve cryptography and rsa. In *Proceedings of the world congress on engineering*, volume 1, pages 419–422.

Manoj, I. V. S. (2010). Cryptography and steganography. *International Journal of Computer Applications*, 1(12):63–68.

Narayan, S. (2018a). A review on elliptic curve cryptography. *International Journal of Emerging Technology and Innovative Engineering*, 4(12).

Narayan, S. (2018b). A review on elliptic curve cryptography. *International Journal of Emerging Technology and Innovative Engineering*, 4(12).

Schroeppel, R., Orman, H., O'Malley, S., and Spatscheck, O. (1995). Fast key exchange with elliptic curve systems. In *Annual International Cryptology Conference*, pages 43–56. Springer.

Silverman, J. and Tate, J. (2015). *Rational Points on Elliptic Curves.*

Smart, N. P. (1999). The discrete logarithm problem on elliptic curves of trace one. *Journal of cryptology*, 12(3):193–196.

Sutikno, S., Surya, A., and Effendi, R. (1998). An implementation of elgamal elliptic curves cryptosystems. In *IEEE. APCCAS 1998. 1998 IEEE Asia-Pacific Conference on Circuits and Systems. Microelectronics and Integrating Systems. Proceedings (Cat. No. 98EX242)*, pages 483–486. IEEE.

Vidhya, E., Sivabalan, S., and Rathipriya, R. (2019). Hybrid key generation for rsa and ecc. In *2019 International Conference on Communication and Electronics Systems (ICCES)*, pages 35–40. IEEE.